

TERM PROJECT – MILESTONE 1

PREDICTIVE ANALYTICS IN HEALTHCARE

BACKGROUND

Diabetes is one of the most common chronic diseases worldwide, and early prediction can greatly improve patient outcomes through timely intervention. As healthcare systems increasingly adopt data-driven solutions, predictive analytics has emerged as a powerful tool to forecast disease risk using patient data. This project focuses on using machine learning to predict diabetes based on key health indicators such as age, BMI, glucose level, and blood pressure. I chose this scenario because of its real-world impact and the personal interest I have in healthcare innovation.

RESEARCH QUESTION / PROBLEM STATEMENT

Main Question: Can we accurately predict the likelihood of a person having diabetes based on key health metrics?

This question is relevant as it aims to reduce the burden of undiagnosed diabetes through proactive identification using machine learning models. Early detection leads to more effective treatment and lifestyle adjustments, ultimately reducing healthcare costs and improving patient outcomes.

PROPOSED GOALS

- Build predictive models to classify whether a person is diabetic or not.
- Compare the performance of multiple classification algorithms.
- Identify key predictors influencing diabetes diagnosis.
- Evaluate model performance using accuracy, confusion matrix, and ROC curve.
- Explore the data to uncover patterns or outliers that may affect the model.

PREVIOUS RESEARCH

A widely cited study by **Smith et al. (2018)** used logistic regression and support vector machines on health datasets to predict diabetes with over 80% accuracy. Their research emphasized the importance of glucose levels and BMI as strong indicators. Similarly, a 2020

research paper by **Fernandes et al.** used ensemble techniques for diabetes prediction, showing improved results compared to traditional models.

DATA COLLECTION

- **Dataset Source:** Kaggle – Diabetes Prediction Accuracy 99%
- **Dataset Size:** Moderate (over 1000 rows, 9 columns)
- **Format:** CSV
- **Citation:** Provided by Adil Ashrafi on Kaggle
- **Quality:** Good, no missing values in the raw dataset, minimal noise
- **Compliance:** The dataset is public and shared under Kaggle’s user-contributed data rules.

DATA CLEANING

- Checked for null values – no missing data found.
- Applied MinMaxScaler for normalization where necessary.
- Removed or treated outliers in columns like “Glucose” and “BMI” using IQR method.
- Renamed columns for clarity and consistency.
- Converted target variable into binary format: 1 = Diabetic, 0 = non-diabetic.

EXPLORATORY DESCRIPTIVE ANALYTICS

We performed the following:

- **Histograms** showed a skew in variables like “Glucose” and “BMI”.
- **Box plots** identified outliers in “Insulin” and “Skin Thickness”.
- **Correlation heatmap** showed strong correlation between “Glucose” and the target variable.
- **Variable Summary:**
 - **Pregnancies** (Ratio): Range 0–17
 - **Glucose** (Ratio): Range 0–199, highly predictive
 - **Blood Pressure** (Ratio): Range 0–122
 - **Skin Thickness** (Ratio): Range 0–99
 - **Insulin** (Ratio): Range 0–846
 - **BMI** (Ratio): Range 0.0–67.1
 - **Diabetes Pedigree Function** (Ratio): Range 0.078–2.42
 - **Age** (Ratio): Range 21–81
 - **Outcome** (Categorical): Binary target variable (0 or 1)

Visualizations support the research question by showing clear differences in feature distribution between diabetic and non-diabetic groups.

PROPOSED SOFTWARE TOOLS

- **Python (Jupyter Notebook)** – Installed locally, also accessible via Google Colab.
- Libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn.
- I've been practicing Python through coursework, online tutorials, and hands-on coding sessions for skill development.

PROPOSED METHODOLOGIES

- **Classification Models:**
 - **Random Forest Classifier**
 - **Logistic Regression**
 - **Bagging Classifier (Ensemble Method)**
- These are suitable for binary classification and can handle nonlinear relationships well.
- Each model will be evaluated on accuracy, precision, recall, F1-score, and ROC-AUC.
- I plan to divide the dataset into 80% training and 20% validation sets.